Machine Learning and AI Methods

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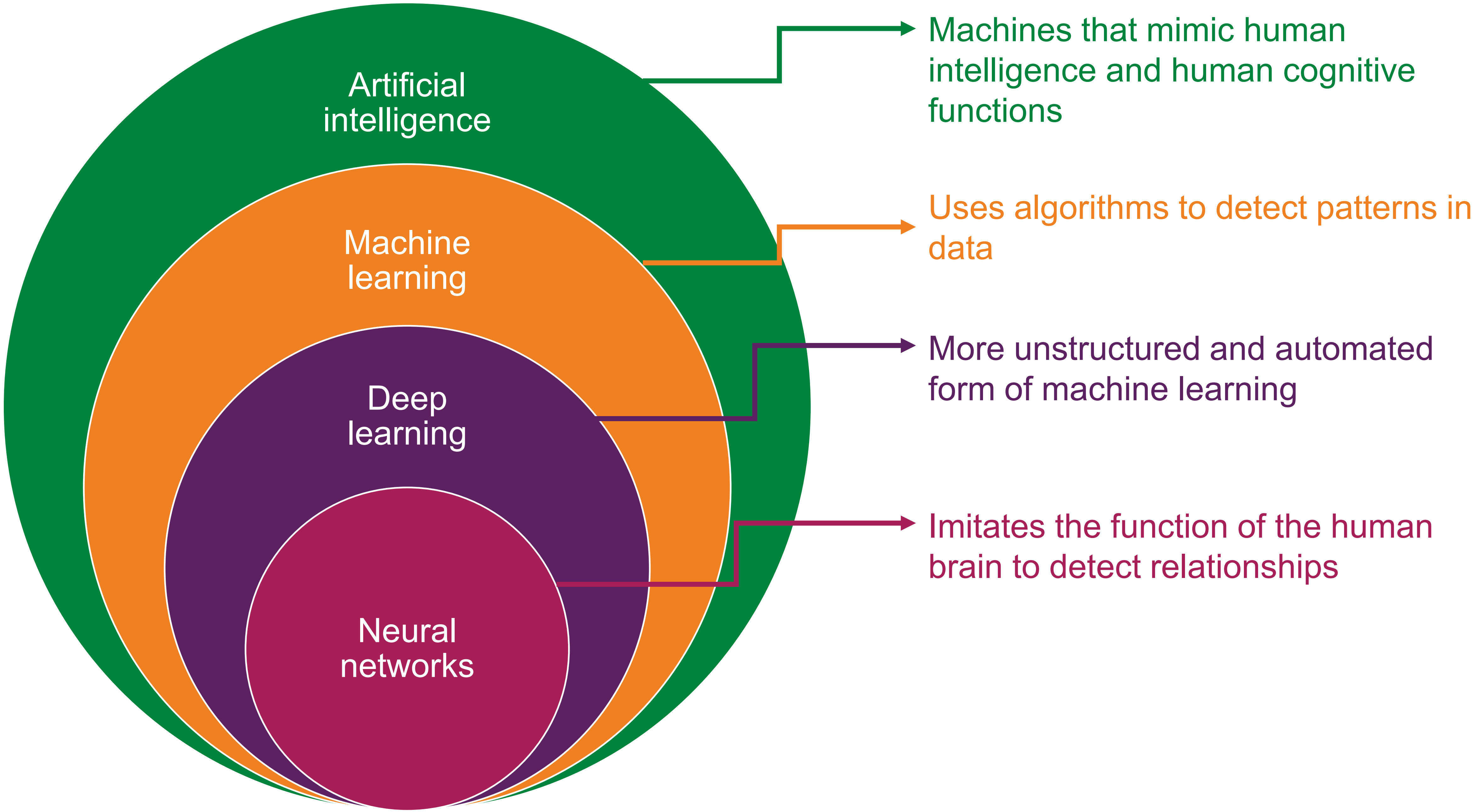

Artificial intelligence
Machine learning
Deep learning
Neural networks
Machines that mimic human intelligence and human cognitive functions
Uses algorithms to detect patterns in data
More unstructured and automated form of machine learning
Imitates the function of the human brain to detect relationships

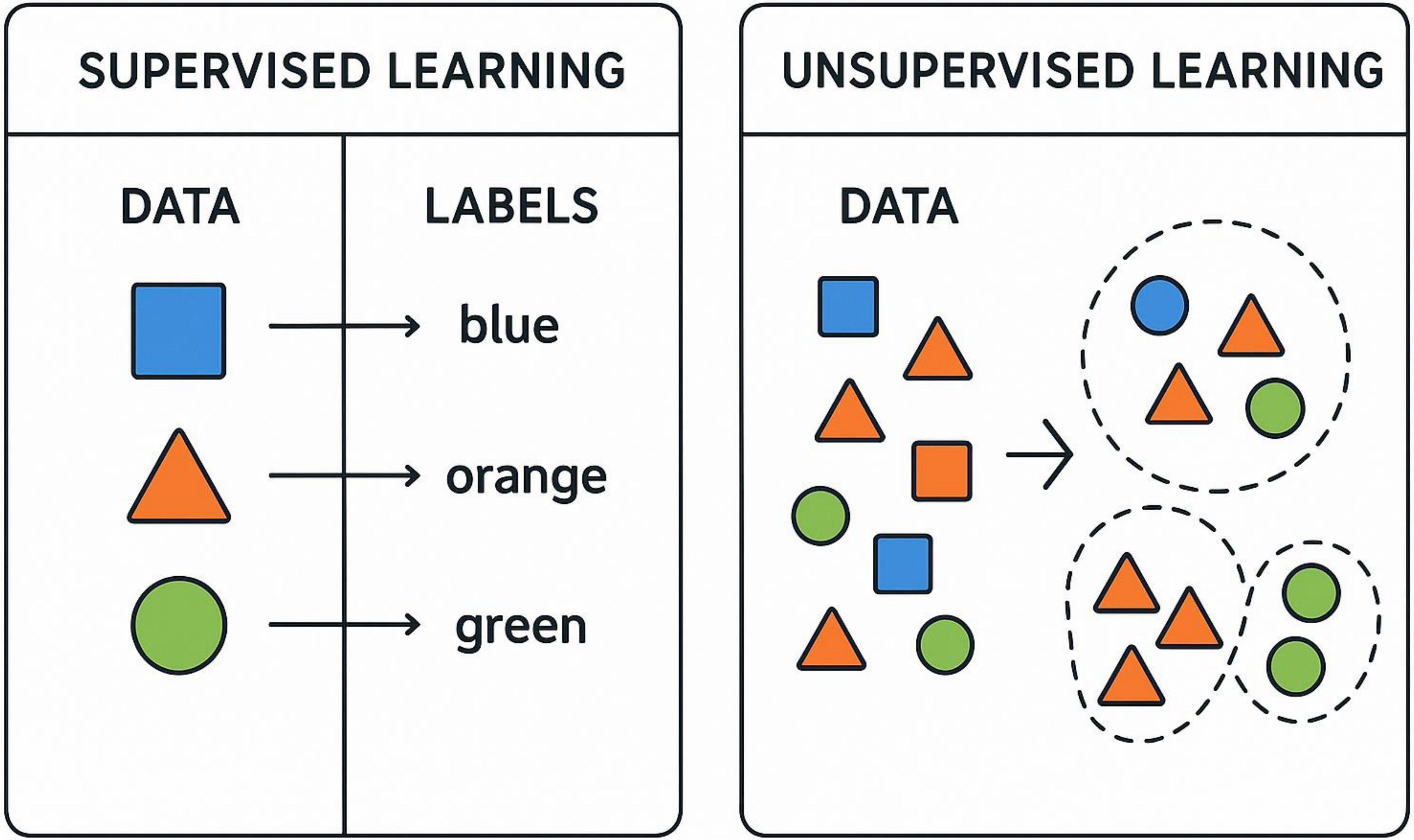
Generative vs predictive AI

Generative AI

- Create something new:
  - Text
  - Image
  - Video
  - Voice
- Unsupervised learning

Predictive AI

- Predict something based on historical data:
  - Diagnosis
  - Prognosis
- Supervised or unsupervised



Example of use of generative AI, image created by ChatGPT

Examples of predictive machine learning algorithms

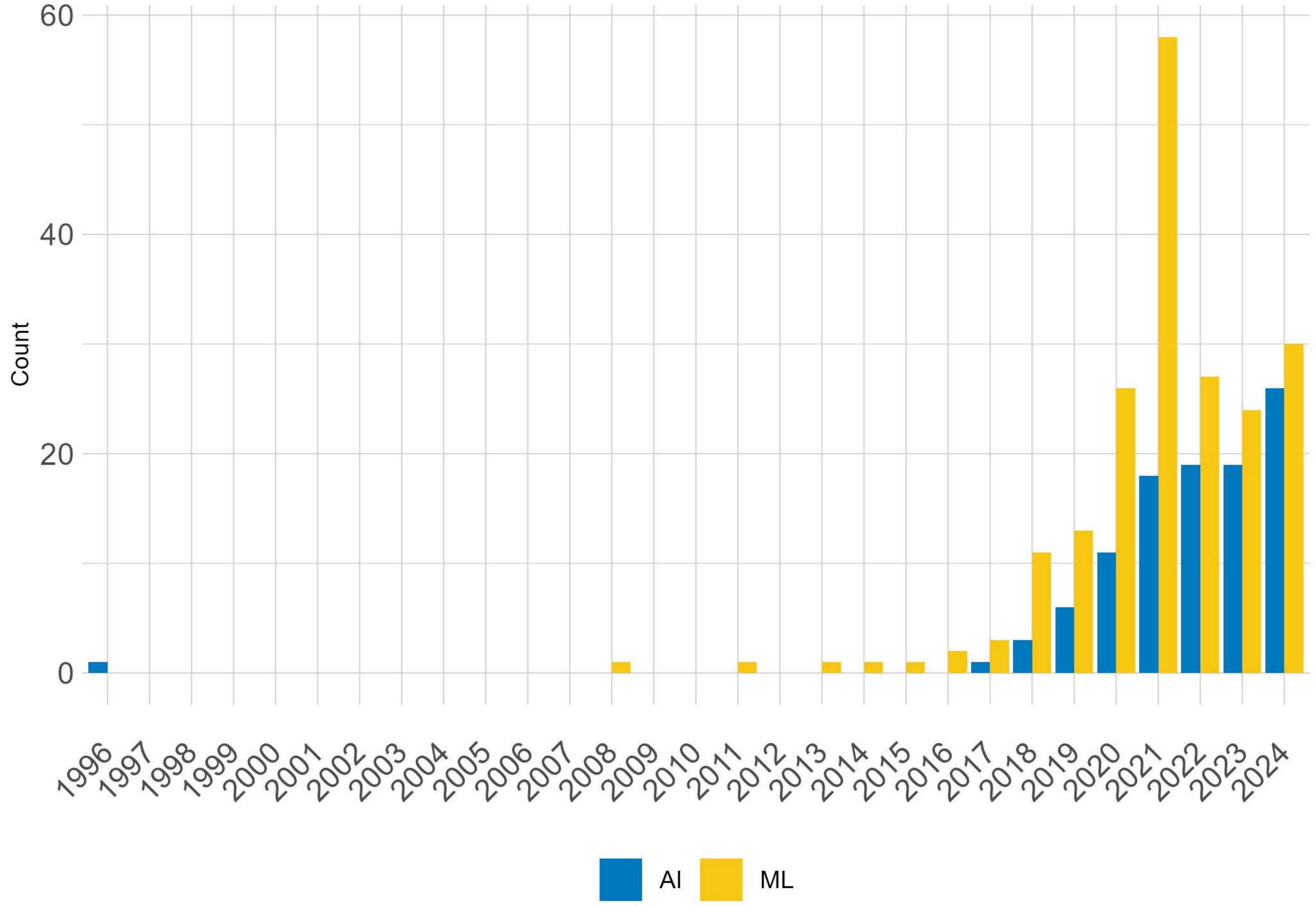
- Linear regression
- Logistic regression
- Decision trees
- Random forests
- Convoluted neural networks
- Support Vector Machines
- Gradient Boosting

Supervised

- K-means clustering
- Hierarchical clustering

Unsupervised

References to AI and ML over time



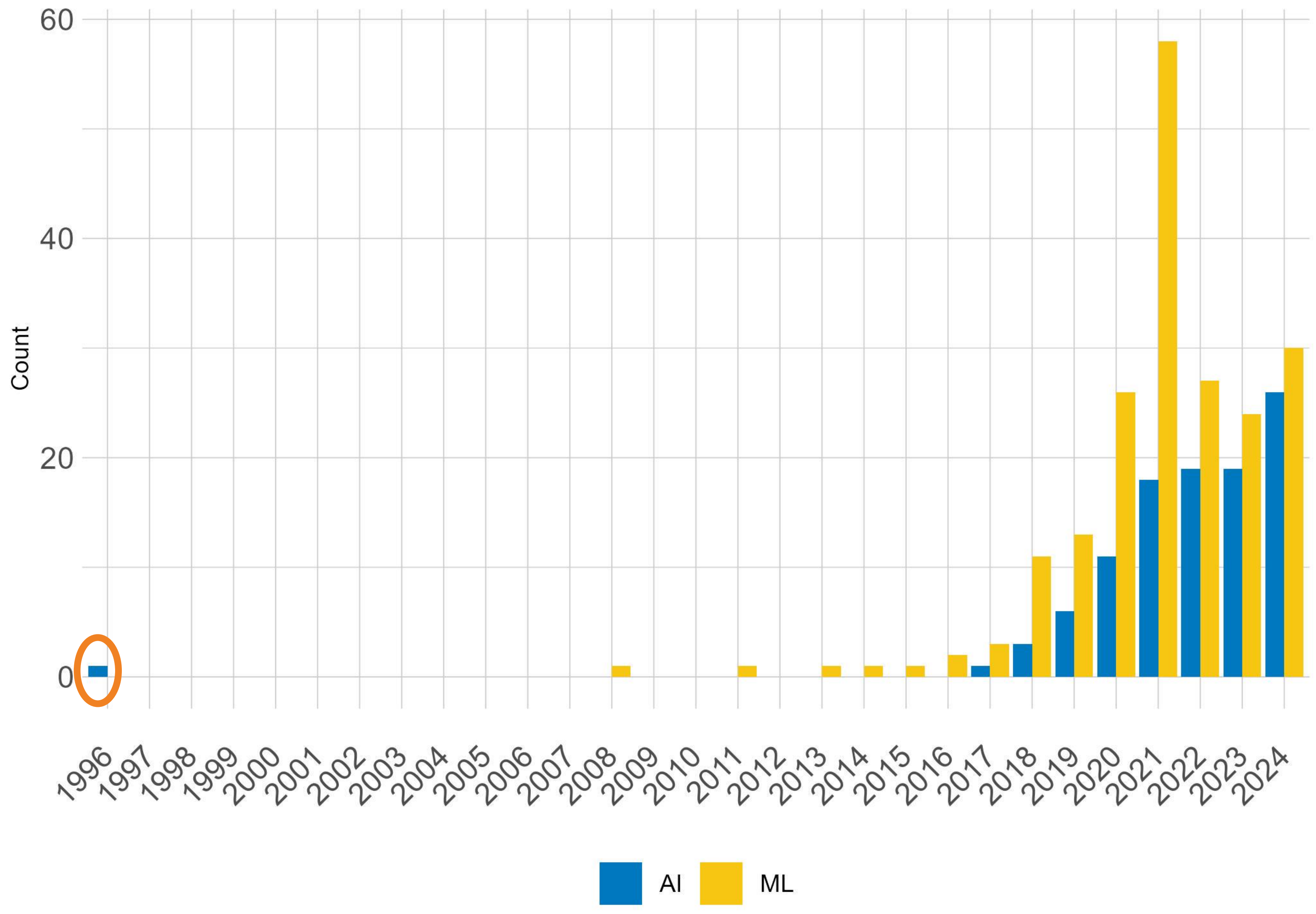
- A (rough) literature review in 3 journals:
  - Am J Ophthalmology
  - Br J Ophthalmology
  - Ophthalmology
- Search terms:
  - Artificial intelligence
  - AI
  - Machine learning
  - ML

Classification = Diagnosis

- The majority of these articles are using AI/ML for diagnosis purposes, a classification problem

- Commonly used classification methods included:
  - Neural networks
  - Deep learning
  - Random forests
  - Logistic regression

An early outlier…



1996: Neural network differentiation of optic neuritis and anterior ischaemic optic neuropathy

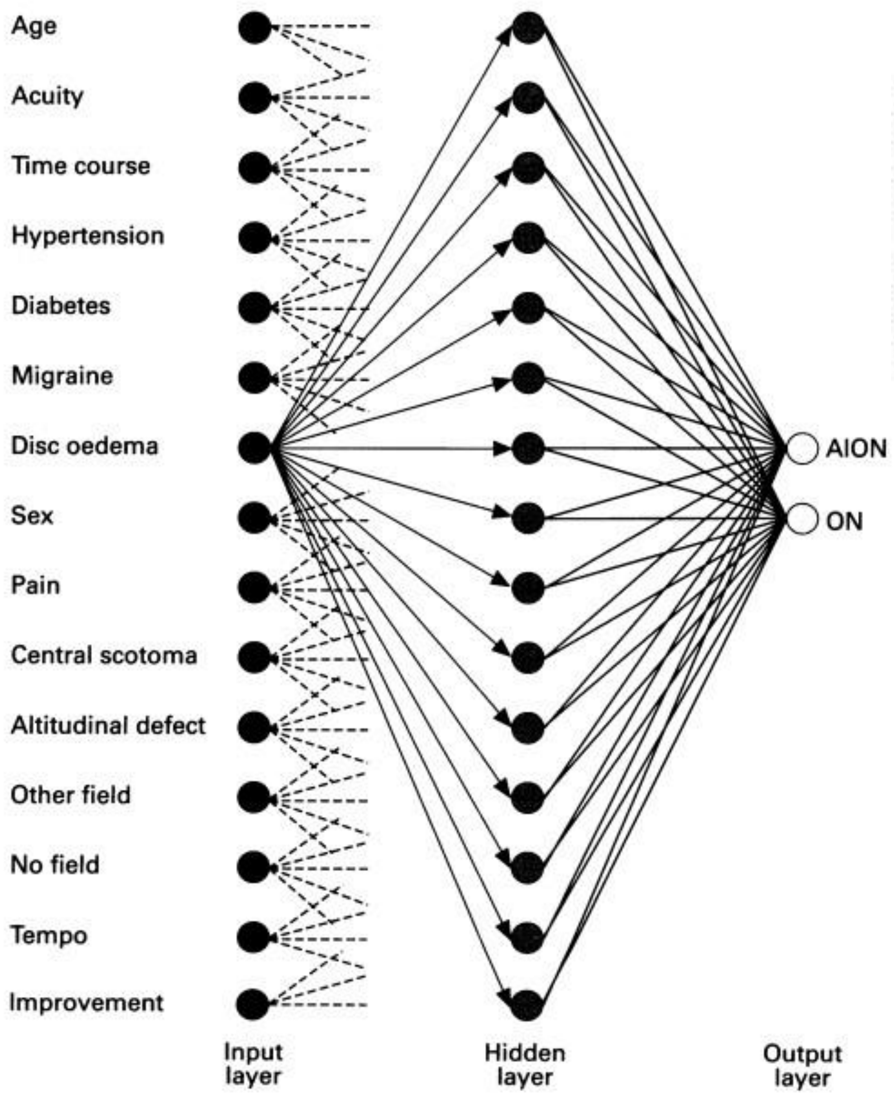
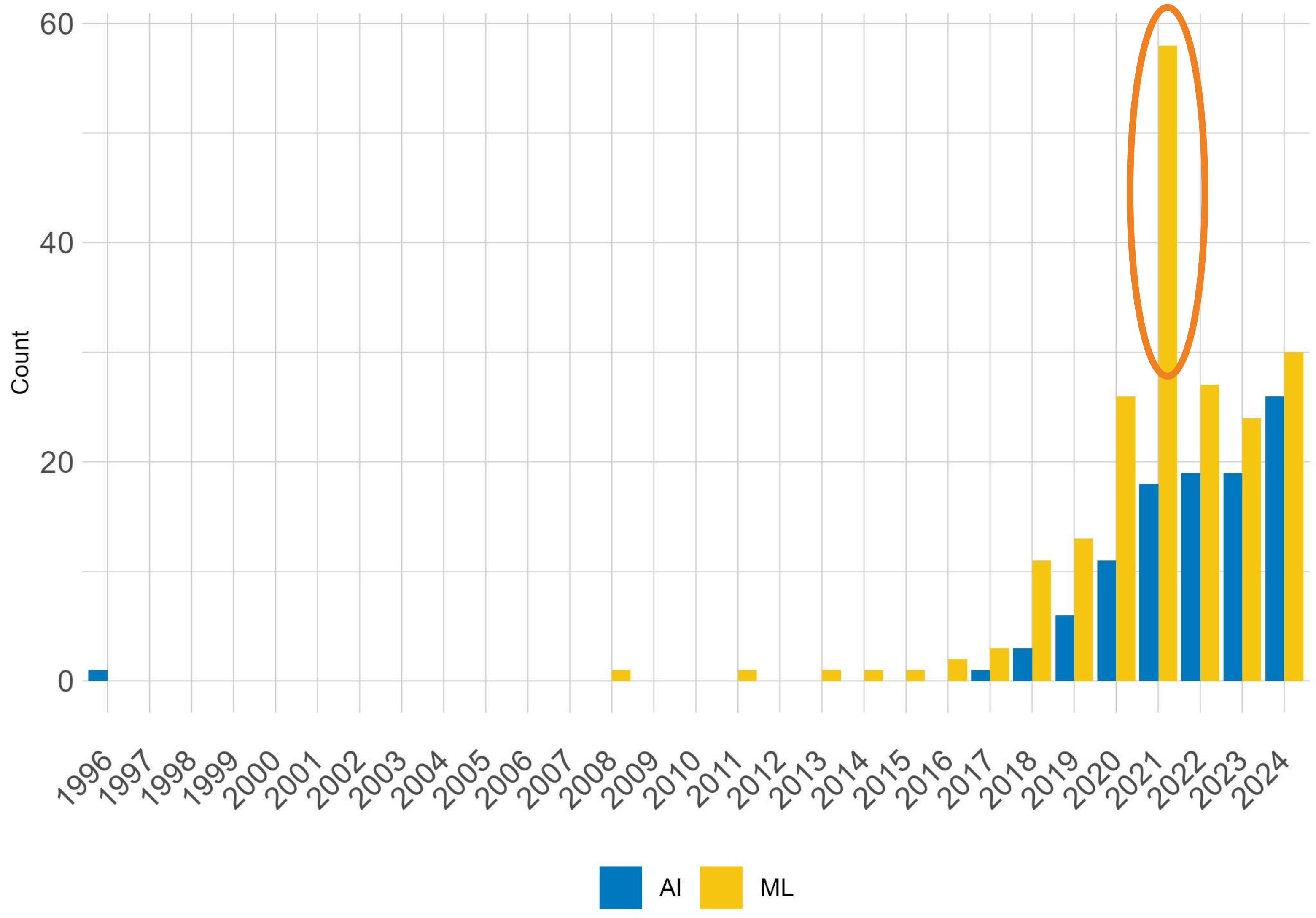


Figure 1 Schematic representation of the neural network. Clinical variables enter from the left and are fed to simulated neurons in the input layer, thus determining their value. The values of the neurons in the hidden layer depend on the values of the input neurons, weighted by the strength of their connections. These weightings are established when the neural network is 'trained'. The values of the output neurons are determined by the values of the hidden layer neurons in a similar fashion. The relative value (whichever is greater) of the output neurons represents the network's preferred diagnosis.

Levin LA, Rizzo JF 3rd, Lessell S. Br J Ophthalmol. 1996 Sep;80(9):835-9.

And a big jump in 2021…



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- Standardization of Uveitis Nomenclature (SUN) Working Group
- 26 articles
- Machine learning using logistic regression
- 85% training/15% validation for each condition
- Goal was classification

Areas for caution

- Only as good as the data they are built on
- Black box – e.g. what happens in the hidden layer?
- Trade-off of complexity versus interpretation

Areas of promise

- Ability to handle large quantities of data
- Automation of complicated procedures/processes
- Better patient care through improved diagnostic and prognostic models

Thank you! Questions?

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